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# Sharpening using Laplacian Mask (8-Neighbour)
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import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload Image
uploaded = files.upload()

# Read uploaded image
filename = list(uploaded.keys())[0]
img = cv2.imread(filename)

# Convert BGR to RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Laplacian Mask with Diagonal Neighbours
laplacian_kernel = np.array([[1, 1, 1],
                             [1,-8, 1],
                             [1, 1, 1]])

# Apply filter
laplacian = cv2.filter2D(img_rgb, -1, laplacian_kernel)

# Sharpen image
sharpened = cv2.subtract(img_rgb, laplacian)

# Display Images
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(np.clip(laplacian,0,255))
plt.title("Laplacian Output")
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(np.clip(sharpened,0,255))
plt.title("Sharpened Image")
plt.axis('off')

plt.show()

# Save Output
cv2.imwrite("Sharpened_Image.jpg", cv2.cvtColor(sharpened, cv2.COLOR_RGB2BGR))

print("Image saved as Sharpened_Image.jpg")
files.download("Sharpened_Image.jpg")
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Saving istockphoto-1598719537-612x612.jpg to istockphoto-1598719537-612x612.jpg
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Original Image

Laplacian Output

Sharpened Image